

1-16題每題4分

In questions 1-4, Determine whether the proposition is a **tautology**(恆真), **contradiction**(矛盾), or **contingency**(偶發). (請直接勾選答案)

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| 1. ☐ Tautology ☐ Contradiction ☒ Contingency | $p \vee q \rightarrow \neg r$ |
| 2. ☒ Tautology ☐ Contradiction ☐ Contingency | $p \vee \neg(p \wedge q)$ |
| 3. ☒ Tautology ☐ Contradiction ☐ Contingency | $(\neg p \wedge q) \rightarrow (p \vee q)$ |
| 4. ☐ Tautology ☒ Contradiction ☐ Contingency | $(\neg q \rightarrow \neg p) \wedge (p \wedge \neg q)$ |

In questions 5–8 suppose the variable **x** represents **students** and **y** represents **courses**, and:

A(y): y is an advanced course

M(y): y is a math course

F(x): x is a freshman

B(x): x is a full-time student

T(x, y): student x is taking course y.

Write the statement using these predicates and any needed quantifiers.

5. Julie is taking MATH281.
6. All students are freshmen.
7. Every freshman is a full-time student.
8. No math course is an advanced course.

Ans:

13. T(Julie, MATH281)

14. $\forall x F(x)$

15. $\forall x (F(x) \rightarrow B(x))$

16. $\forall y [M(y) \rightarrow \neg A(y)]$.

In questions 9-12, Determine whether the statement is a **proposition** or not. (請直接勾選答案)

9. ☒Yes ☐No The Earth is flat.
10. ☒Yes ☐No $2 + 2 = 4$
11. ☐Yes ☒No $x + 5 = 10$
12. ☒Yes ☐No The sky is blue.

In questions 13–16 suppose **P(x, y)** is a predicate and the **universe for the variables x and y is {1, 2, 3}**.

Suppose P(1, 1), P(1, 2), P(2, 1), P(2, 2), P(3, 2), P(3, 3) are true, and P(x, y) is false otherwise. Determine whether the following statements are **True or False**. (請直接勾選答案)

13. ☐True ☒False $\exists x \forall y P(x, y)$.
14. ☐True ☒False $\neg \exists x \exists y (P(x, y) \wedge \neg P(y, x))$.
15. ☒True ☐False $\forall y \exists x (P(x, y) \rightarrow P(y, x))$
16. ☐True ☒False $\forall x \forall y (x \neq y \rightarrow (P(x, y) \vee P(y, x)))$.

Ans: 列出P(x,y)的真偽值比較容易看

y	1	2	3
x			
1	T	T	F
2	T	T	F
3	F	T	T

5. False, 沒有任一列是TTT

6. False, 反例 $x=3, y=2$

7. True, $\forall y \exists x x=y$

8. False, 反例 $x=3, y=1$

17. (8%) Simplify the following compound proposition to minimize the number of " $\wedge$ ", " $\vee$ ", and " $\neg$ " operations:

$$(p \wedge q) \vee (\neg p \wedge \neg r \wedge \neg s) \vee (\neg p \wedge \neg q \wedge s) \vee (\neg p \wedge q \wedge s) \vee (\neg p \wedge r \wedge \neg s)$$

Ans: 原式 $\equiv (p \wedge q) \vee (\neg p \wedge \neg r \wedge \neg s) \vee (\neg p \wedge s) \vee (\neg p \wedge r \wedge \neg s)$
 $\equiv (p \wedge q) \vee (\neg p \wedge \neg s) \vee (\neg p \wedge s) \equiv (p \wedge q) \vee \neg p \equiv q \vee \neg p$

18. Show that $p \rightarrow (q \vee r)$ and $\neg p \vee (\neg r \rightarrow q)$ are **equivalent**

- (a) (8%) using a truth table.
 (b) (8%) using logical equivalences

Ans: (a) As shown in the following truth table, the truth values of $p \rightarrow (q \vee r)$ and $\neg p \vee (\neg r \rightarrow q)$ are all the same for all possible combination of the truth values of p, q , and r , it follows that $p \rightarrow (q \vee r)$ and $\neg p \vee (\neg r \rightarrow q)$ are equivalent.

p	$\neg p$	q	r	$\neg r$	$(q \vee r)$	$\neg r \rightarrow q$	$p \rightarrow (q \vee r)$	$\neg p \vee (\neg r \rightarrow q)$
T	F	T	T	F	T	T	T	T
T	F	T	F	T	T	T	T	T
T	F	F	T	F	T	T	T	T
T	F	F	F	T	F	F	F	F
F	T	T	T	F	T	T	T	T
F	T	T	F	T	T	T	T	T
F	T	F	T	F	T	T	T	T
F	T	F	F	T	F	F	T	T

Ans: (b)

$$\begin{aligned}
 & p \rightarrow (q \vee r) \\
 \equiv & \neg p \vee (q \vee r) && (\text{by } p \rightarrow q \equiv \neg p \vee q) \\
 \equiv & \neg p \vee (\neg q \rightarrow r) && (\text{by } p \rightarrow q \equiv \neg p \vee q) \\
 \equiv & \neg p \vee (\neg r \rightarrow q) && (\text{by } p \rightarrow q \equiv \neg q \rightarrow \neg p)
 \end{aligned}$$

19. (10%) Prove the proposition: "A positive integer n is even if $3n^2+8$ is even."

Ans: 根據proof by contraposition, 若能證明『若 n 為奇數, 則 $3n^2+8$ 必為奇數』則原題中的命題成立.

若 n 為奇數, 則 $n = 2a+1$, $a \in \mathbb{Z}$

$$\Rightarrow 3n^2+8 = 3(2a+1)^2 + 8 = 3(4a^2 + 4a + 1) + 8 = 2(6a^2 + 6a + 5) + 1 = 2k+1, \text{ 其中 } k=6a^2 + 6a + 5$$

$\Rightarrow 3n^2+8$ 為奇數.

因此可證若 $3n^2+8$ 為偶數, 則 n 亦為偶數.

20. (10%) Prove that $\sqrt{3}$ is irrational (無理數).

Ans: 證明 $\sqrt{3}$ 為無理數的方法是採用反證法。首先假設 $\sqrt{3}$ 是有理數, 可寫成兩個互質的整數 a/b 的比值, 即 $\sqrt{3} = a/b$ 。將此式平方後得到 $3 = a^2/b^2$, 即 $a^2 = 3b^2$ 。這表示 a^2 是3的倍數, 因此 a 也必須是3的倍數。若 $a = 3k$ (k 為整數), 則代入 $a^2 = 3b^2$ 可得 $(3k)^2 = 3b^2$, 即 $9k^2 = 3b^2$, 進而得到 $3k^2 = b^2$ 。這表示 b^2 也是3的倍數, 因此 b 也必須是3的倍數。這樣就得出 a 和 b 都是3的倍數, 與它們互質的假設相矛盾。因此, 原來的假設是錯誤的, $\sqrt{3}$ 不可能是有理數, 必為無理數。